

THE VARIANCE OF SQUAREFREE INTEGERS IN SHORT INTERVALS OF LENGTH UP TO $X^{4/7}$

ABSTRACT. Gorodetsky, Matomäki, Radziwiłł and Rodgers proved that the number of squarefree integers in $(x, x+H]$, for x chosen at random in $[X, 2X]$, has variance asymptotic to $C\sqrt{H}$ when $H \leq X^{6/11-\varepsilon}$. We extend the range to $H \leq X^{4/7-\varepsilon}$. Only one step of their argument changes: in the contribution of integers with a large square divisor, the large value estimate of Guth and Maynard replaces Huxley's. At $H = X^{4/7}$ several independent constraints of the argument become sharp at once, and we show that, within this method and with the large value and moment estimates we tried, $4/7$ cannot be passed unless the range of the estimate for small square divisors is enlarged.

1. INTRODUCTION

Let μ be the Möbius function, so that μ^2 is the indicator function of the squarefree integers. For $X \geq 2$ and $1 \leq H \leq X$ put

$$\mathcal{V}(X, H) := \frac{1}{X} \int_X^{2X} \left| \sum_{x < n \leq x+H} \mu^2(n) - \frac{6}{\pi^2} H \right|^2 dx.$$

The variance $\mathcal{V}(X, H)$ is of order $\sqrt{H}$, much smaller than the order H that a random model with independent integers would give: the condition $p^2 \nmid n$ correlates the squarefree integers in an interval. Hall [Ha] proved the asymptotic below for $H \leq X^{2/9-\varepsilon}$; see also [GMaR] for the higher moments. Let

$$C := \frac{\zeta(3/2)}{\pi} \prod_p \left(1 - \frac{3}{p^2} + \frac{2}{p^3} \right).$$

Theorem A (Gorodetsky–Matomäki–Radziwiłł–Rodgers) ([GMRR, Theorem 1]). *Let $\varepsilon \in (0, \frac{1}{100})$, $X \geq 2$ and $H \leq X^{6/11-\varepsilon}$. Then $\mathcal{V}(X, H) = C\sqrt{H} + O_\varepsilon(H^{1/2-\varepsilon/16})$.*

Under the Lindelöf hypothesis the same authors reach $H \leq X^{2/3-\varepsilon}$, and they show that the Riemann hypothesis is equivalent to the bound $\mathcal{V}(X, H) \ll H^{1/2+\varepsilon}$ throughout $H \leq X^{1-\varepsilon}$ [GMRR, Theorems 1 and 3]. Our result is the following.

Theorem 1. *Let $\varepsilon \in (0, \frac{1}{100})$, $X \geq 2$ and $H \leq X^{4/7-\varepsilon}$. Then*

$$\mathcal{V}(X, H) = C\sqrt{H} + O_\varepsilon(H^{1/2-\varepsilon/64}).$$

Since $6/11 = 0.5454\dots$ and $4/7 = 0.5714\dots$, Theorem 1 enlarges the unconditional range by the factor $X^{2/77}$.

Where the exponent comes from. The argument of [GMRR] writes $\mu^2(n) = \sum_{d^2|n} \mu(d)$ and splits at a parameter z . The square divisors $d^2 \leq z$ produce the main term $C\sqrt{H}$; this part is proved for $z \leq \min\{X/H^{1/2+\varepsilon}, H^{1/2-\varepsilon}X^{1/2}\}$ (Proposition 2 below). The square divisors $d^2 > z$ contribute only to the error term, and this is proved for $z \geq H^{4/3+\varepsilon}$ (Proposition 3). The two ranges overlap exactly when $H^{4/3} \leq X/H^{1/2}$, that is, when $H \leq X^{6/11}$.

The exponent $4/3$ comes from Dirichlet polynomials $M(1 + 2it) = \sum_{d \sim D} \mu(d) d^{-1-2it}$ at heights $T \approx X/H$. The critical level set is $|M(1 + 2it)| \approx D^{-1/4}$: in the normalisation of Guth and Maynard this is a polynomial of length $N = D$ taking values of size $N^{3/4}$. This is precisely the regime in which their large value estimate [GM, Theorem 1.1] improves on the Montgomery–Halász–Huxley bound. Gorodetsky, Matomäki, Radziwiłł and Rodgers point out [GMRR, end of Section 2] that, at $H = X^{6/11}$, the only case of the Heath-Brown decomposition of $\mu(d)$ they could not handle is the one in which $\mu(d)$ is replaced by two smooth sums of equal length; this is the same configuration. Inserting [GM, Theorem 1.1] lowers $4/3$ to $5/4$ (Proposition 8), and $H^{5/4} \leq X/H^{1/2}$ gives $H \leq X^{4/7}$.

Why $4/7$. At $H = X^{4/7}$, with $z = X/H^{1/2}$ and $T = X/H$, the polynomial length is $D = \sqrt{z} = T^{5/6}$, and several constraints become sharp at once (Section 6): the upper end $X/H^{1/2}$ of the range of Proposition 2; the Weyl bound in the large- d estimate (the source of the restriction $H \leq X^{4/7}$ already present in Proposition 3); and both power-saving terms of [GM, Theorem 1.1]. The Weyl bound can be replaced by Bourgain’s, and for $D \geq T^{5/6}$ the Guth–Maynard estimate for long polynomials [GM, Proposition 12.1] replaces Theorem 1.1 there; but that replacement is blocked at the same point: at the level $\sigma = 3/4$ no large value estimate we know gives fewer than $T^{1/2}$ points, and for $T^{1/2}$ unit intervals the fourth and twelfth moments of ζ give the same bound. Thus, within this method (which treats M as a polynomial with generic bounded coefficients), any further progress requires enlarging the range of Proposition 2, the part of the argument that produces the main term.

Arithmetic progressions. [GMRR, Theorem 2] is the analogue of Theorem A for progressions to a prime modulus $q \geq x^{5/11+\varepsilon}$. The same substitution, with a hybrid (character) form of the Guth–Maynard estimate, should give $q \geq x^{3/7+\varepsilon}$; see Remark 10. We have not checked this in detail.

2. THE ARGUMENT OF GORODETSKY, MATOMÄKI, RADZIWIŁŁ AND RODGERS

For $z \geq 1$ that is not the square of an integer, $x \in [X, 2X]$ and $H \leq X$, put

$$\Sigma_{\leq z}(x) := \sum_{\substack{x < nd^2 \leq x+H \\ d^2 \leq z}} \mu(d) - H \sum_{d^2 \leq z} \frac{\mu(d)}{d^2}, \quad \Sigma_{> z}(x) := \sum_{\substack{x < nd^2 \leq x+H \\ d^2 > z}} \mu(d) - H \sum_{z < d^2 \leq 2X} \frac{\mu(d)}{d^2}.$$

(The requirement that z not be a square only removes the ambiguity between $d^2 < z$ and $d^2 \leq z$ in [GMRR].) Since $\sum_d \mu(d) d^{-2} = 6/\pi^2$, for every $x \in [X, 2X]$

$$(1) \quad \sum_{x < n \leq x+H} \mu^2(n) - \frac{6}{\pi^2} H = \Sigma_{\leq z}(x) + \Sigma_{> z}(x) - H \sum_{d^2 > 2X} \frac{\mu(d)}{d^2}, \quad \left| H \sum_{d^2 > 2X} \frac{\mu(d)}{d^2} \right| \leq \frac{2H}{\sqrt{X}}.$$

Proposition 2 ([GMRR, Proposition 1]). *Let $\varepsilon \in (0, \frac{1}{100})$, $X \geq 1$, $X^\varepsilon \leq H \leq X^{2/3-\varepsilon}$ and $H^{1+\varepsilon} \leq z \leq \min\{X/H^{1/2+\varepsilon}, H^{1/2-\varepsilon} X^{1/2}\}$. Then, as $X \rightarrow \infty$,*

$$\frac{1}{X} \int_X^{2X} |\Sigma_{\leq z}(x)|^2 dx = C\sqrt{H} + O(H^{1/2-\varepsilon/10}).$$

Proposition 3 ([GMRR, Proposition 2]). *Let $\varepsilon \in (0, \frac{1}{100})$, $X \geq 1$, $X^\varepsilon \leq H \leq X^{4/7-\varepsilon}$ and $z \geq H^{4/3+\varepsilon}$. Then $\frac{1}{X} \int_X^{2X} |\Sigma_{> z}(x)|^2 dx \ll H^{1/2-\varepsilon/8}$.*

We do not change Proposition 2. We replace Proposition 3 by Proposition 8 below, keeping the first half of its proof, which we now recall. Let $D \geq 1$ be a power of 2 and put

$$\mathcal{D} := \{d \in (D, 2D] : d^2 > z\}, \quad M(s) = M_D(s) := \sum_{d \in \mathcal{D}} \frac{\mu(d)}{d^s}.$$

and

$$\Sigma_{>z}^{(D)}(x) := \sum_{\substack{x < nd^2 \leq x+H \\ d \in \mathcal{D}}} \mu(d) - H \sum_{d \in \mathcal{D}} \frac{\mu(d)}{d^2}.$$

Let $\mathbf{D}$ be the set of the $O(\log X)$ powers of two D with $\sqrt{z}/2 < D \leq \sqrt{3X}$. Every d with $z < d^2 \leq 3X$ lies in exactly one $\mathcal{D}$, $D \in \mathbf{D}$, and $nd^2 \leq x + H \leq 3X$, so

$$(2) \quad \Sigma_{>z}(x) - \sum_{D \in \mathbf{D}} \Sigma_{>z}^{(D)}(x) = H \sum_{\substack{d^2 > 2X \\ d \in \bigcup_{D \in \mathbf{D}} \mathcal{D}}} \frac{\mu(d)}{d^2} = O\left(\frac{H}{\sqrt{X}}\right).$$

The Dirichlet series of $n \mapsto \sum_{d \in \mathcal{D}, d^2 | n} \mu(d)$ is $\zeta(s)M(2s)$. Perron's formula, a shift of the contour to $\Re s = 1/2$ (the pole at $s = 1$ produces the subtracted term $H \sum_{d \in \mathcal{D}} \mu(d)d^{-2}$), Plancherel's theorem and a dyadic decomposition of the t -integral give the following.

Lemma 4 ([GMRR, Section 5, up to display (40)]). *Let $\varepsilon \in (0, \frac{1}{100})$ and $X^\varepsilon \leq H \leq X^{4/7-\varepsilon}$. Uniformly in D and z ,*

$$\frac{1}{X} \int_X^{2X} |\Sigma_{>z}^{(D)}(x)|^2 dx \ll H \sup_{X/H < T < X^2} \frac{1}{T} \int_{|t| < T} |\zeta(\tfrac{1}{2} + it)M_D(1 + 2it)|^2 dt + 1.$$

In [GMRR] the integral is split according to the level sets

$$S(V) := \{t \in [-T, T] : V \leq |M(1 + 2it)| < 2V\}, \quad V = 2^k D^{-1/2} \leq 1,$$

and Huxley's large value estimate [Hu] gives

$$|S(V)| \ll (V^{-2} + T \min\{D^{-1}V^{-2}, D^{-2}V^{-6}\})(\log 2X)^6.$$

When the term V^{-2} dominates, the Weyl bound gives the contribution $\ll HT^{-2/3+o(1)}$, which is acceptable when $H \leq X^{4/7-\varepsilon}$. Otherwise Cauchy–Schwarz and the fourth moment give

$$\frac{H}{T} V^2 \int_{S(V)} |\zeta(\tfrac{1}{2} + it)|^2 dt \ll HV^2 \min\{D^{-1/2}V^{-1}, D^{-1}V^{-3}\}(\log X)^5 \leq HD^{-3/4}(\log X)^5,$$

with equality of the two terms at $V = D^{-1/4}$; this requires $D \geq H^{2/3+}$, that is $z \geq H^{4/3+}$.

3. INPUTS

Lemma 5 (Guth–Maynard [GM, Theorem 1.1]). *Let (b_n) be complex numbers with $|b_n| \leq 1$, and let $(t_r)_{r \leq R}$ be 1-separated points of $[0, T]$ with $|\sum_{n=N}^{2N} b_n n^{it_r}| \geq W$ for all r . Then for every $\eta > 0$, if T is large enough in terms of η ,*

$$R \leq T^\eta (N^2 W^{-2} + N^{18/5} W^{-4} + TN^{12/5} W^{-4}).$$

We use the theorem as stated in the arXiv version (v2, 7 April 2026) of [GM]. For comparison, the Mean Value Theorem and the Montgomery–Halász–Huxley estimate give $R \leq T^{o(1)}(N^2 W^{-2} + T \min\{N W^{-2}, N^4 W^{-6}\})$; at $W = N^{3/4}$ this is $N^{1/2} + TN^{-1/2}$, while Lemma 5 gives $N^{3/5} + TN^{-3/5}$.

Lemma 6 (Weyl bound, [Ti, Theorem 5.12]). $\zeta(\tfrac{1}{2} + it) \ll (|t| + 2)^{1/6} \log(|t| + 2)$ for real t .

Lemma 7 (Ingham's fourth moment [In]). $\int_{-T}^T |\zeta(\frac{1}{2} + it)|^4 dt \ll T \log^4 T$ for $T \geq 2$.

4. THE LARGE SQUARE DIVISORS

Proposition 8. Let $\varepsilon \in (0, \frac{1}{100})$, $X \geq 2$, $X^\varepsilon \leq H \leq X^{4/7-\varepsilon}$ and $z \geq H^{5/4+\varepsilon}$, where z is not the square of an integer. Then

$$\frac{1}{X} \int_X^{2X} |\Sigma_{>z}(x)|^2 dx \ll_\varepsilon H^{1/2-\varepsilon/8}.$$

The only difference from Proposition 3 is the exponent $5/4$ in place of $4/3$.

Lemma 9. Let $D \geq 4$ be a power of 2, $T \geq 2$, $0 < V \leq 1$, and let $S(V)$ be as above. For every $\eta > 0$, if T is large enough in terms of η , the Lebesgue measure of $S(V)$ satisfies

$$|S(V)| \leq 4(2T)^\eta (V^{-2} + D^{-2/5}V^{-4} + 2TD^{-8/5}V^{-4}).$$

Proof. Let $\{t_r\}$ be a maximal 1-separated subset of $S(V)$. By maximality $S(V) \subset \bigcup_r (t_r - 1, t_r + 1)$, so $|S(V)| \leq 2\#\{t_r\}$. Put $N = D$ and $b_n = \mu(n)D/n$ for $n \in \mathcal{D}$, $b_n = 0$ for the other $n \in [N, 2N]$, so that $|b_n| \leq 1$ and $\sum_{n=N}^{2N} b_n n^{-2it} = DM(1 + 2it)$. The points $2t_r$ with $t_r < 0$ are 2-separated in $[0, 2T]$ after the change $t \mapsto -t$ and satisfy $|\sum_n b_n n^{i(2|t_r|)}| \geq DV$; the points $2t_r$ with $t_r \geq 0$ satisfy the same with b_n replaced by $\bar{b}_n$. Lemma 5 with T replaced by $2T$ and $W = DV$, applied to each of the two families, gives

$$\#\{t_r\} \leq 2(2T)^\eta (D^2(DV)^{-2} + D^{18/5}(DV)^{-4} + 2TD^{12/5}(DV)^{-4}),$$

which is the claim. $\square$

Proof of Proposition 8. By (2) and Cauchy–Schwarz over the $O(\log X)$ values $D \in \mathbf{D}$ (the right side of (2) has square $\ll H^2/X \leq H^{1/2-1/7}$), it suffices to show that for each $D \in \mathbf{D}$ and each $T \in (X/H, X^2)$,

$$(3) \quad \frac{H}{T} \int_{|t| < T} |\zeta(\frac{1}{2} + it)M(1 + 2it)|^2 dt \ll_{\varepsilon, \eta} X^{3\eta} H^{1/2-2\varepsilon/5}$$

for every small $\eta > 0$; Lemma 4 then gives $\frac{1}{X} \int_X^{2X} |\Sigma_{>z}|^2 \ll X^{3\eta} (\log X)^2 H^{1/2-2\varepsilon/5}$, and since $X \leq H^{1/\varepsilon}$, the choice $\eta = \varepsilon^2/20$ gives the bound $H^{1/2-\varepsilon/4+o(1)} \ll H^{1/2-\varepsilon/8}$. We may assume X large.

We record the sizes of the parameters. Since $D > \sqrt{z}/2 \geq H^{5/8+\varepsilon/2}/2$, $T > X/H$, $X^\varepsilon \leq H \leq X^{4/7-\varepsilon}$ and $H \leq X$, we have

$$(4) \quad D^{-4/5} \leq 2H^{-\frac{1}{2}-\frac{2\varepsilon}{5}}, \quad H^{\frac{7}{6}}X^{-\frac{2}{3}} \leq X^{-\frac{7\varepsilon}{6}} \leq H^{-\frac{7\varepsilon}{6}}, \quad H^{\frac{7}{8}}X^{-\frac{1}{2}} \leq X^{-\frac{7\varepsilon}{8}} \leq H^{-\frac{7\varepsilon}{8}}.$$

Also $|M(1 + 2it)| \leq \sum_{D < d \leq 2D} d^{-1} < 1$.

Small values. On the set where $|M(1 + 2it)| < D^{-1/2}$ the integrand is at most $D^{-1}|\zeta(\frac{1}{2} + it)|^2$, and Lemma 7 with Cauchy–Schwarz gives the contribution $\ll HD^{-1} \log^2 T \ll H^{3/8} \log^2 X$.

Level sets. It remains to bound, for each of the $O(\log X)$ values of $V = 2^k D^{-1/2} \leq 1$,

$$B(V) := \frac{H}{T} (2V)^2 \int_{S(V)} |\zeta(\frac{1}{2} + it)|^2 dt.$$

By Lemma 9, $|S(V)| \ll X^{2\eta} \max\{V^{-2}, D^{-2/5}V^{-4}, TD^{-8/5}V^{-4}\}$. We use two bounds for the ζ -integral:

$$(5) \quad \int_{S(V)} |\zeta(\tfrac{1}{2} + it)|^2 dt \leq |S(V)| \max_{|t| \leq T} |\zeta(\tfrac{1}{2} + it)|^2 \ll |S(V)| T^{1/3} \log^2 T,$$

$$(6) \quad \int_{S(V)} |\zeta(\tfrac{1}{2} + it)|^2 dt \leq |S(V)|^{1/2} \left(\int_{-T}^T |\zeta(\tfrac{1}{2} + it)|^4 dt \right)^{1/2} \ll |S(V)|^{1/2} T^{1/2} \log^2 T,$$

by Lemmas 6 and 7.

Case 1: the maximum is V^{-2} . By (5), $B(V) \ll X^{3\eta} HT^{-2/3} \leq X^{3\eta} H^{5/3} X^{-2/3} = X^{3\eta} H^{1/2} \cdot H^{7/6} X^{-2/3} \leq X^{3\eta} H^{1/2-7\epsilon/6}$.

Case 2: the maximum is $D^{-2/5}V^{-4}$. By (6),

$$B(V) \ll X^{3\eta} \frac{H}{T} V^2 D^{-1/5} V^{-2} T^{1/2} = X^{3\eta} H D^{-1/5} T^{-1/2} \leq X^{3\eta} H^{3/2} X^{-1/2} D^{-1/5}.$$

Since $D^{-1/5} \leq 2H^{-1/8-\epsilon/10}$, this is $\ll X^{3\eta} H^{1/2} \cdot H^{7/8} X^{-1/2} \cdot H^{-\epsilon/10} \leq X^{3\eta} H^{1/2-7\epsilon/8}$.

Case 3: the maximum is $TD^{-8/5}V^{-4}$. By (6),

$$B(V) \ll X^{3\eta} \frac{H}{T} V^2 (TD^{-8/5}V^{-4})^{1/2} T^{1/2} = X^{3\eta} H D^{-4/5} \leq 2X^{3\eta} H^{1/2-2\epsilon/5}.$$

This is the step where [GMRR] obtain $HD^{-3/4}$. Note that V cancels: the bound holds uniformly over the level sets, including the critical one $V \approx D^{-1/4}$.

Summing over V and adding the small values proves (3) (after renaming η to absorb the logarithms). $\square$

5. PROOF OF THEOREM 1

If $H \leq X^{6/11-\epsilon}$, Theorem 1 follows from Theorem A since $H^{1/2-\epsilon/16} \leq H^{1/2-\epsilon/64}$. Suppose that $X^{6/11-\epsilon} < H \leq X^{4/7-\epsilon}$; in particular $X^\epsilon \leq H$. Put $\epsilon_1 := \epsilon/4$ and choose $z \in [H^{5/4+\epsilon_1}, 2H^{5/4+\epsilon_1}]$ that is not the square of an integer.

Proposition 2 applies with ϵ_1 . We have $X^{\epsilon_1} \leq H \leq X^{2/3-\epsilon_1}$ and $z \geq H^{1+\epsilon_1}$. Next, $z \leq X/H^{1/2+\epsilon_1}$ follows from $2H^{7/4+2\epsilon_1} \leq X$, which holds for large X because $(4/7-\epsilon)(7/4+\epsilon/2) = 1 - \frac{41}{28}\epsilon - \frac{\epsilon^2}{2} < 1$. Finally $z \leq H^{1/2-\epsilon_1} X^{1/2}$ follows from $2H^{3/4+2\epsilon_1} \leq X^{1/2}$, which holds because $\frac{4}{7}(\frac{3}{4} + \frac{\epsilon}{2}) < \frac{1}{2}$.

Proposition 8 applies with ϵ_1 , since $X^{\epsilon_1} \leq H \leq X^{4/7-\epsilon_1}$ and $z \geq H^{5/4+\epsilon_1}$.

Write $\|f\|^2 := \frac{1}{X} \int_X^{2X} |f(x)|^2 dx$. By (1) and the two propositions,

$$\begin{aligned} \|\Sigma_{\leq z}\|^2 &= C\sqrt{H} + O(H^{1/2-\epsilon/40}), & \|\Sigma_{> z}\|^2 &\ll H^{1/2-\epsilon/32}, \\ \left| H \sum_{d^2 > 2X} \frac{\mu(d)}{d^2} \right|^2 &\leq \frac{4H^2}{X} \leq 4H^{1/2-1/7}, \end{aligned}$$

where the last step uses $H^{3/2} \leq X^{6/7}$. Writing $E := \Sigma_{> z} - H \sum_{d^2 > 2X} \mu(d)d^{-2}$, we get $\|E\|^2 \ll H^{1/2-\epsilon/32}$ and

$$\mathcal{V}(X, H) = \|\Sigma_{\leq z} + E\|^2 = \|\Sigma_{\leq z}\|^2 + O(\|\Sigma_{\leq z}\| \|E\| + \|E\|^2) = C\sqrt{H} + O(H^{1/2-\epsilon/64}).$$

This proves Theorem 1. $\square$

6. WHY THE METHOD STOPS AT $4/7$

Write $H = X^h$, $D = X^\delta$, $T = X^\tau$ and $V = D^{-v}$ with $0 \leq v \leq \frac{1}{2}$. In Lemma 4 the weight of the frequencies $|t| \asymp T$ is $H \min\{HT/X, X/(HT)\}$ (this is what the proof in [GMRR] gives before taking the supremum), so the critical height is $T = X/H$, i.e. $\tau = 1 - h$. Proposition 2 allows δ as small as $\frac{1}{2} - \frac{h}{4}$ (that is, $z = X/H^{1/2}$), and no smaller. We need, for all $\delta \geq \frac{1}{2} - \frac{h}{4}$ and all v , that

$$H \cdot \frac{1}{T} V^2 \int_{S(V)} |\zeta(\frac{1}{2} + it)|^2 dt \ll H^{1/2-c}.$$

At $h = 4/7$ and $\delta = \frac{1}{2} - \frac{h}{4} = \frac{5}{14}$ (so $D = T^{5/6}$), each of the following is sharp:

- (1) the range $z \leq X/H^{1/2}$ of Proposition 2;
- (2) Case 1 above, with the Weyl bound: $HT^{-2/3} \leq H^{1/2}$ iff $h \leq 4/7$;
- (3) Case 2: $H^{3/2}X^{-1/2}D^{-1/5} \leq H^{1/2}$ at $D = H^{5/8}$ iff $h \leq 4/7$;
- (4) Case 3: $HD^{-4/5} \leq H^{1/2}$ iff $D \geq H^{5/8}$, i.e. $z \geq H^{5/4}$, and $H^{5/4} \leq X/H^{1/2}$ iff $h \leq 4/7$.

Item (2) can be relaxed: Bourgain's bound $\zeta(\frac{1}{2} + it) \ll |t|^{13/84+o(1)}$ [Bo] moves it to $h \leq 29/50$. For $h > 4/7$ the length $D = X^{1/2-h/4}$ exceeds $T^{5/6}$, so the dominant term of [GM, Theorem 1.1] there is $N^{18/5}W^{-4}$, that is item (3), and for $T^{5/6} \leq N \leq T$ Guth and Maynard's estimate for long polynomials [GM, Proposition 12.1] is a complete replacement for Theorem 1.1 (for $\sigma \geq 7/10$). This replacement is itself blocked at $4/7$, at the same place as (1) and (4), for the following reason. At the level $\sigma = 3/4$ (that is, $v = 1/4$, $W = N^{3/4}$) and $N = D \in [T^{5/6}, T]$, every large value estimate we checked (the Mean Value Theorem, Huxley, Jutila's estimate with $k = 3$ [Ju], [GM, Theorem 1.1 and Proposition 12.1]) gives at best $R \lesssim T^{1/2}$ (we have not gone through the full compilation of large value estimates in [TTY]); this is attained by [GM, Proposition 12.1], whose term $T^{1/2}N^{3-4\sigma}$ equals $T^{1/2}$ at $\sigma = 3/4$. (For $4/7 \leq h \leq 2/3$ the smallest length allowed by Proposition 2, $D = X^{1/2-h/4}$, does lie in $[T^{5/6}, T]$.) For a set of $R = T^{1/2}$ unit intervals the fourth moment and Heath-Brown's twelfth moment [HB] give the same bound $T^{3/4}$ for $\int_S |\zeta(\frac{1}{2} + it)|^2$, and the pointwise bound is worse. The level $\sigma = 3/4$ therefore contributes $HD^{-1/2}T^{-1/4}$, which is $\leq H^{1/2}$ only if $D \geq H^{3/2}X^{-1/2}$, i.e. $\delta \geq \frac{3h-1}{2}$. Since $\frac{3h-1}{2} \leq \frac{1}{2} - \frac{h}{4}$ if and only if $h \leq \frac{4}{7}$, for every $h > 4/7$ the large- d estimate, with these inputs, requires z beyond the range of Proposition 2. This argument treats M as a Dirichlet polynomial with arbitrary coefficients bounded by 1 and uses only the size of $S(V)$ and moments of ζ ; it does not rule out gains from the arithmetic structure of μ (for instance via Heath-Brown's identity, as discussed in [GMRR, Section 2]).

We confirmed this numerically by optimising over all (δ, v) with the full set of inputs (Mean Value Theorem, Huxley, [GM, Theorem 1.1 and Proposition 12.1] with the infimum over k , Jutila with $k = 3$, Bourgain's exponent, and the fourth and twelfth moments), at the critical height $T = X/H$; larger T do not change the values. The smallest admissible δ (accurate to about 10^{-4} , the grid resolution) is shown below; it exceeds $\frac{1}{2} - \frac{h}{4}$ for every $h > 4/7$ we tested. For $h \geq 0.585$ no δ works, because Case 1 fails for every D once $h > 29/50$, even with Bourgain's bound.

h	$\frac{1}{2} - \frac{h}{4}$ (Prop. 2)	smallest admissible δ	$\frac{3h-1}{2}$
0.5715	0.35713	0.35726	0.35725
0.5750	0.35625	0.36346	0.36250
0.5800	0.35500	0.37172	0.37000
0.5850	0.35375	none	0.37750

Consequently, the next step is an extension of Proposition 2 to z somewhat larger than $X/H^{1/2}$. In [GMRR] this bound arises in the off-diagonal terms of the variance of the small- d part, after the Fourier expansion of the sawtooth function, where pairs $n_1 d_2^2 \neq n_2 d_1^2$ are counted with a lemma on quadratic irrationals. Under the Lindelöf hypothesis the large- d part only requires $z \geq H^{1+\varepsilon}$, and then $H \leq X^{2/3}$ is the limit of Proposition 2 itself. With Proposition 2 extended, Bourgain’s exponent would allow up to $h < 29/50$ in the present argument.

Remark 10 (Arithmetic progressions). [GMRR, Theorem 2] gives the analogue of Theorem A for progressions modulo a prime $q \geq x^{5/11+\varepsilon}$. It is deduced from [GMRR, Proposition 3], valid for $z \leq x^{-\varepsilon}\sqrt{qx}$, and [GMRR, Proposition 4], which requires $z \geq (x/q)^{4/3+\varepsilon}$ and is proved with the hybrid form of Huxley’s estimate, the hybrid fourth moment, and the Petrow–Young Weyl bound for Dirichlet L -functions [PY]; Proposition 4 already carries the assumption $q \geq x^{3/7+\varepsilon}$, the analogue of $H \leq X^{4/7-\varepsilon}$. A hybrid estimate of the form $R \lesssim N^2 W^{-2} + N^{18/5} W^{-4} + qTN^{12/5} W^{-4}$ would give, by the same computation, $z \geq (x/q)^{5/4+\varepsilon}$ and hence $q \geq x^{3/7+\varepsilon}$ (since $(x/q)^{5/4} \leq \sqrt{qx}$ if and only if $q \geq x^{3/7}$). Chen [Ch, Theorem 1.3] proves a hybrid Guth–Maynard estimate that agrees with this shape at $N = (qT)^{5/6}$, which is the critical length when $q = x^{3/7}$; a check of the exponents with Chen’s bounds for $N \in [(qT)^{3/4}, (qT)^{5/6}]$ and $N \geq (qT)^{5/6}$, at $T = x^{o(1)}$, again gives $q \geq x^{3/7}$. We have not verified the reduction in [GMRR, Proposition 4] in detail, and [Ch] is a recent preprint.

PROVENANCE

This note was produced with Claude (Anthropic) at the request of Siddharth Iyer, as part of a search for problems from the work of Matomäki. The literature search behind it (arXiv and the web, October 2026) found no earlier claim of the range $H \leq X^{4/7-\varepsilon}$, but it was not exhaustive. The argument has not been refereed.

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